

Is Dark Matter the only explanation for Galaxy Rotation Curves?

U2210176, 127

The existence of dark matter is seen as a natural interpretation of the anomalous trend shown by galaxy rotation curves (GRCs). We reproduce the GRC of the Milky Way using the tangent point method (TPM) for the radial range of $4.5 < R < 8$ kpc. Our results confirm previous research by showing a steady velocity profile and galactic mass increasing linearly with galactocentric radius. This conflicts with established Newtonian dynamics. We therefore highlight the possibility of Newton's Laws being inadequate to explain galactic dynamics as an alternative to dark matter. We also discuss empirical evidence to support our case.

1 Introduction

Galaxy rotation curves (GRCs) provide one of the strongest pieces of evidence for dark matter while offering great insights into its properties and distribution [1]. GRC refers to the plot of mean rotational velocity V_{rot} around the galactic centre as a function of radius R . The rotational velocity of the galaxy has been mapped using several methods such as CO, HII, and HI emission observations [2]. We use HI observations in our experiment to map the rotation of the Milky Way. Hydrogen in its atomic state is present abundantly in the interstellar medium (ISM). Magnetic interaction induced by the spin alignment of the atom's

electron and proton causes hyperfine splitting of its ground state, with the parallel spin configuration having slightly higher energy than the antiparallel one. The energy gap is small enough for the spin-flip transition to occur despite the cold temperatures of the ISM. Therefore, clouds of HI constantly emit microwave photons that possess energy equivalent to this gap, corresponding to radiation of wavelength 21-cm and frequency 1.4204058 GHz as given by $E = h\nu$ [3]. The 21-cm emission appears as a broad peak on the HI-cloud spectrum that is Doppler-shifted due to the Earth's relative motion. This shift can be translated into a measure of radial velocity v_R as follows:

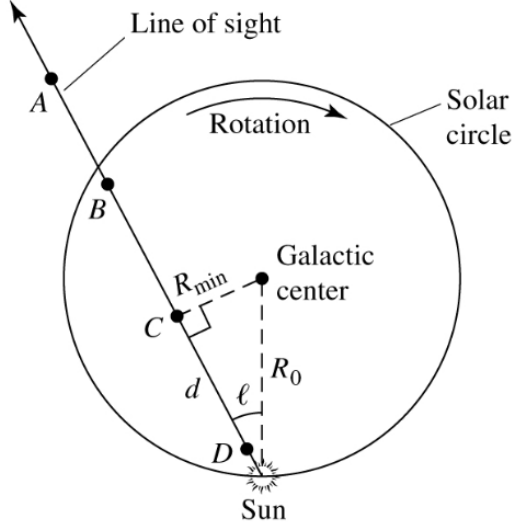


Figure 1: Geometry of the tangent point method. $R_{\min}$ is the galactocentric distance R to an HI cloud at tangent point C. R_0 is the galactocentric radius of the Sun. l denotes the galactic longitude that moves within the solar circle to map all the tangential points of galactic disks with radius $R_{\min}$. Adapted from [4].

$$\nu' = \nu_0 \left(1 - \frac{v_R}{c}\right) \quad (1)$$

Here, ν_0 is the rest frequency, ν' the observed frequency, c is the speed of light. v_R is the velocity component along the line of sight (positive for a receding source from the Earth). In this experiment, we assume a circular galactic disc, which allows us to exploit the geometry of a circle to make our observations.

Fig. 1 shows a locus of points equidistant from the galactic center at $R_0 = 7.9 \pm 0.4$ kpc defined as the solar circle. The galactic disks of radii $0 < R < R_0$ inside the solar circle have tangential points at which V_{rot} is entirely parallel to the line of sight, giving maximum radial velocity. This allows us to write V_{rot} as a function of $v_r(\text{max})$ [2]:

$$V_{\text{rot}}(R) = v_r(\text{max}) + V_0 \sin l \quad (2)$$

Here, $V_0 = V(R_0)$. We have chosen $(7.9 \pm 0.4 \text{ kpc}, 220 \pm 13 \text{ km s}^{-1})$ as our galactic constants [5]. As per the above relation, maximum v_r corresponds to a minimum frequency $\nu_{\min}'$. We therefore require $\nu_{\min}'$ to find the velocities at the tangential points.

Gauss's Law of Gravity dictates that only the enclosed mass within the galactic disk of radius R will affect the orbital speed at R [6]. By equating gravitational and centripetal accelerations, we can write the enclosed mass M as a function of R and compare the mass-radius profile to theoretical expectations:

$$\frac{GM(R)}{R^2} = \frac{V_{\text{rot}}(R)^2}{R} \quad (3)$$

Here, G is the gravitational constant. Given the decline in the luminous matter at large galactocentric radii, we expect V_{rot} to decrease as per the above relation (Fig. 2). Consequently, we expect the $M(R)$ profile to stay constant for radii beyond the central bulge which has a high concentration of luminous matter relative to the disk. Any increase in mass beyond $R > 3 \text{ kpc}$ will be negligible compared to this mass (Fig. 6).

We find however that the results are at odds with the above theory as outlined in Section 3. We discuss the implications of these results in Section 4 and propose some

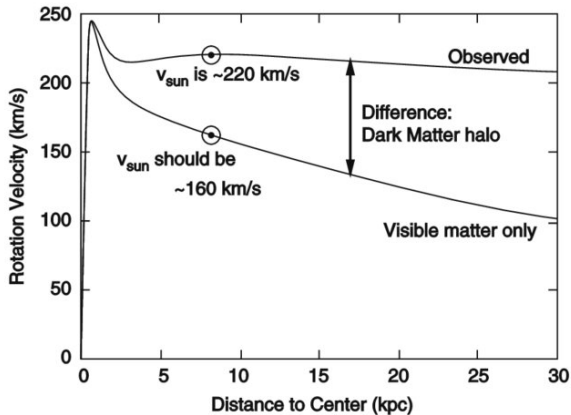


Figure 2: The bottom curve shows the theoretical GRC considering only the existence of luminous matter. The top curve shows the true GRC as observed through experimentation. This discrepancy can either be explained by the existence of unseen matter or the inadequacy of Newtonian dynamics at galactic scales. Adapted from [7].

improvements that can be made to the experiment in Section 5.

2 Methodology

The imperfect axisymmetry of the Milky Way galactic bar causes perturbations in the velocity field that result in unreliable estimates using TPM. These effects are less pronounced moving further from the central bulge ($R > 4.5$ kpc) [2]. We therefore constrained the galactic longitude range to $35^\circ < l < 90^\circ$ for our experiment to minimise any perturbing effects. We chose the Local Standard of Rest (LSR) as the reference point, which although has the same galactocentric radius and mean V_{rot} as the Sun, executes a non-eccentric orbit around the galactic centre, making relative observations easier [8].

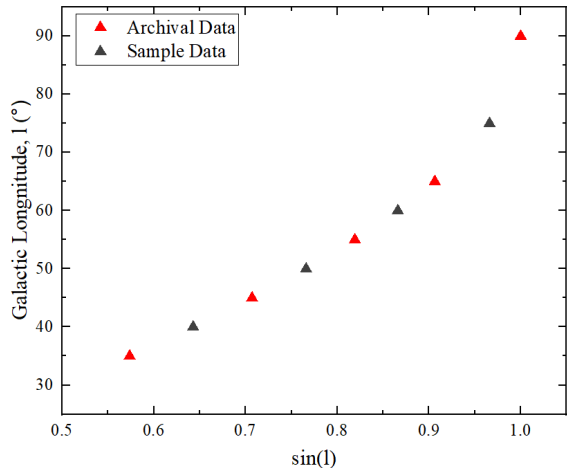


Figure 3: Analysing optimum galactic longitudes for observation by identifying the gaps in measurements. The red triangles represent the archival data, while the dark triangles represent new observations taken to produce varied and adequately spaced results.

We used the 2.3m-diameter radio telescopes, with the diffraction limit given by $\theta = \frac{1.22\lambda}{D}$. This works out at 6.382° . As per the Nyquist sampling criterion [9], we spaced our sample observations within the $3.191^\circ < \Delta l < 6.382^\circ$ range. We combined our observations with the archival data provided for longitudes $l = 45^\circ, 55^\circ, 65^\circ$ and 90° . To ensure an adequately spaced sampling that covers a varied range of radii, we plotted the galactic longitudes against $\sin l$ to identify the gaps in the measurements (Fig. 3). The ideal longitudes for observations were identified as $l = 40^\circ, 50^\circ, 60^\circ$ and 75° .

The observed spectra included radio interference peaks that were masked to isolate the 21-cm radiation line. We observed a broad peak extending over a range of frequencies, all corresponding to the red-

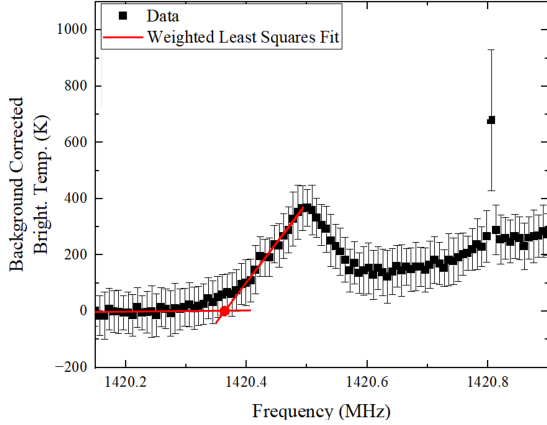


Figure 4: A background-corrected HI spectrum for $l = 75^\circ$. The peak rises from a redshifted frequency of about 1420.4 MHz. $\nu_{\min}'$ is determined by the x-intercept (red dot) of the background (modelled as the red $y=0$ line) and the linear section of the emission peak (steep red line).

shifted 21-cm radiation since the spectra of multiple gas clouds were being analysed along the line of sight. The farther the gas cloud, the more redshifted the frequency is expected to be. We eliminated the background noise by making a linear fit to the flat section of the spectrum and subtracting it from the brightness temperature. We found it reasonable to look for $\nu_{\min}'$ in the region where the 21-cm emission peak begins to rise from the flat background signal. We modelled the background signal as a $y=0$ line that intersected with the approximately linear section of the emission peak. The x-intercept of the two linearly modelled curves gave us the $\nu_{\min}'$ (Fig. 4). This strategy was repeated for each of the longitudes sampled.

We expect the error bars on $\nu_{\min}'$ to be dominated by the errors on the background

signal rather than the 21-cm emission line. This is because a flat line gives a larger variation when translated vertically than a steep slope. The standard errors on the background signal therefore indicated the maximum possible displacement the signal could make, providing a measure of uncertainty on $\nu_{\min}'$. Sources of random errors include incorrect telescope alignment on the sky and arbitrary determination of the rising of the peak against the background due to different curve shapes which may introduce observational bias. This means that the true rising points of the peaks may be higher than the determined x-intercept. The random errors are however dwarfed by the systematic uncertainties in the measurements of R_0 and V_0 . With measurements of $\nu_{\min}'$ at our disposal, we used Eq. 1 to determine v_r at each galactic longitude. We corrected v_r for the relative motion of the LSR. Finally, Eq. 2 was used to obtain the GRC of the Milky Way.

3 Results

The galactic longitudes were translated to values for galactocentric radius R using $R = R_0 \sin l$, whereas V_{rot} was obtained using Eq. 2. We used these values to plot the GRC of the Milky Way as shown in Fig. 5. The curve shows a steady V_{rot} centred at 235 km s^{-1} . Furthermore, the shape of our GRC closely resembles the results by Fich

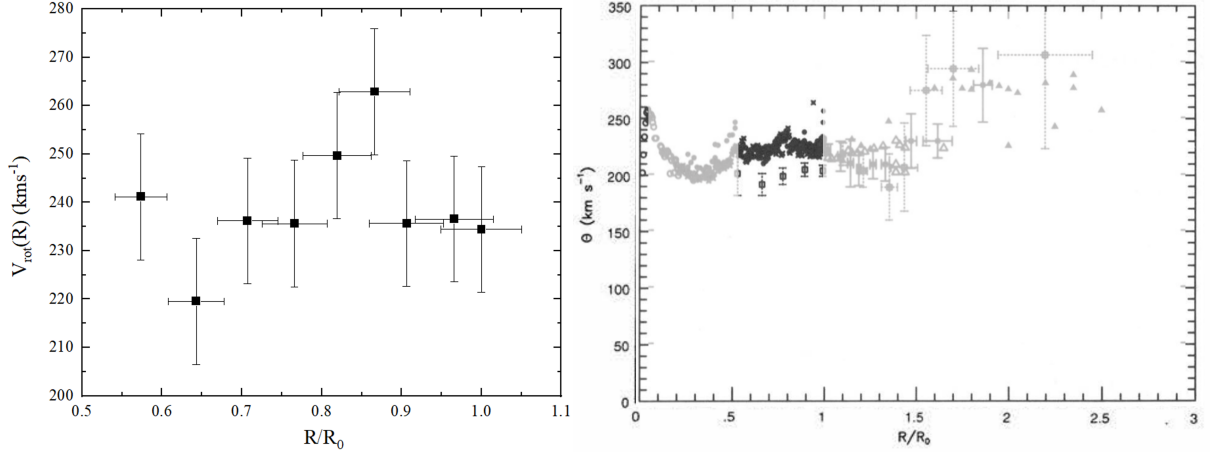


Figure 5: The GRC produced from our observations (left) is compared with other GRCs (right) for the radial range $4.5 < R < 8$ kpc (highlighted section on the right) [10]. The shape of the two GRCs is very similar, although vertically offset due to systematics. The GRCs conflict with expected results, giving a steady velocity-profile instead of one that declines at large radii.

et al. (1989) who used the same range of galactocentric radii [10]. However, our results share fewer similarities with extended GRCs that account for the outer regions of the galaxies [11]. This may be due to the large error bars on points beyond the solar circle dominating the shape of the weighted curve fit. Our results are therefore uninfluenced by the error bars on points much further away and show the true V_{rot} .

Fig. 6 shows a plot of enclosed galactic mass M as a function of galactocentric radius R . The mass is seen to rise approximately linearly as we move out from the galactic centre. The mass-radius profile can be used to estimate the distribution of surface mass density (SMD). Assuming a spherical or disk-shaped topology, the SMD is inversely proportional to R^2 or R respectively, indicating that in either case, the mass is unevenly distributed.

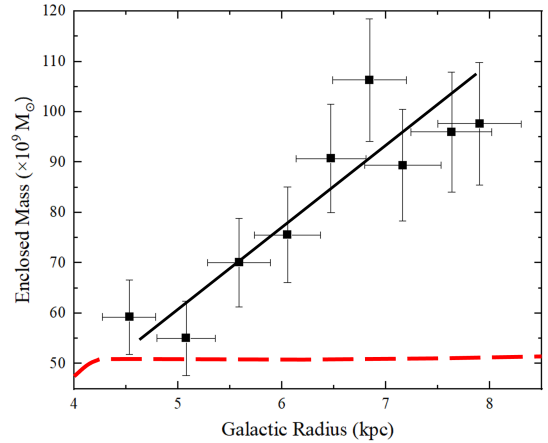


Figure 6: The scatter plot shows the galactic mass profile obtained from the GRC. The enclosed mass increases approximately linearly (indicated by the solid black line) in contrast with the expected curve (indicated by the dashed red line).

4 Discussion

We expected V_{rot} at $R = 8$ kpc to be equal to $V_0 = 220 \text{ km s}^{-1}$. However, a mean value of 235 km s^{-1} suggests a translation of data points induced by systematic errors. These originate from the uncertainties on the R_0 and V_0 values taken from literature. It

may also result from underestimating $\nu_{\min}$, using the intercept method, resulting in a higher V_{rot} . The interpretation of the data however remains unaffected since we are concerned with the shape of the velocity profile. A considerable decrease in the visible matter is observed as we move out from the highly luminous centre of the Milky Way. With a decrease in the mass-to-radius ratio, the laws of Newtonian gravity dictate that the rotational velocities at galactocentric radius R should also decline. This expectation conflicts with experimental observation which provides a relatively constant velocity profile. The mass-radius profile (Fig. 6) derived from the GRC shows the enclosed mass increasing linearly with R , an anomalous trend under the expectation that it should remain approximately constant relative to the mass in the central bulge.

Two conclusions can be derived from the data at hand. A more natural conclusion is the existence of invisible mass that counteracts the diminishing effect of increasing radius on V_{rot} to sustain Newtonian dynamics. This unseen mass exerts a sufficiently strong gravitational influence to alter galactic kinematics but does not interact with the electromagnetic spectrum [12]. A second possible conclusion is that the laws of classical Newtonian mechanics are inadequate to explain galactic dynamics and therefore require modifications. Mod-

ified Newtonian Dynamics (MOND) proposed by Mordehai Milgrom in 1983 revises Newton's law of gravity by suggesting a quadratic relationship between force and acceleration instead of one that is linear [13]. This relation holds for objects possessing acceleration less than the experimentally calculated value of $a_0 = 1.2 \times 10^{-10} \text{ ms}^{-2}$ [14]. For accelerations $>a_0$, Newtonian dynamics are restored. Low acceleration is a characteristic feature of matter in galactic halos. Therefore, MOND works well for matter at the perimeter of galaxies, while reproducing Newtonian kinematics for the inner regions [14].

5 Conclusion

We used the HI 21-cm emission line observations coupled with the tangent point method to map the rotation of the Milky Way using $(7.9 \pm 0.4 \text{ kpc}, 220 \pm 13 \text{ kms}^{-1})$ as our galactic constants. We showed that TPM produces an accurate depiction of the true galaxy rotation curve for the radial range of $4.5 < R < 8 \text{ kpc}$ and its findings are consistent with the Λ CDM cosmological model. However, TPM is limited to a narrow range of radial distances, rendering it unable to map the complete GRC of the Milky Way and provide accurate mass distribution models. Other methods need to be employed such as measuring high-resolution spectra of Halo K Giants [15].

Dark matter is not the only explanation for the steady velocity-profile of the Milky Way. MOND provides a commendable solution for the anomalous galactic dynamics backed by observational evidence. A study done in 2022 to compare the CDM and the MOND framework concluded that the GRCs are better explained by MOND [16]. Furthermore, the mass discrepancy-acceleration relation proposed by MOND shows that for accelerations $>a_0$, dark matter is not required to sustain Newtonian mechanics [17, 18].

Despite the strong case for MOND, independent experiments that provide compelling evidence for the existence of dark matter cannot be ignored. These include anisotropies in the Cosmic Microwave Background that can only be explained by the existence of a source of gravity other than baryonic matter [19]. Observation of the Bullet Cluster using gravitational lensing shows the mass distribution to be offset from its expected location, providing yet another evidence for dark matter [20]. Moreover, the anomalous galactic dynamics might just be a consequence of the yet unexplored properties of dark matter.

References

- [1] J. F. Navarro, The Astrophysical Journal (1998).
- [2] Chemin, Laurent, Renaud, Florent, and Soubiran, Caroline, AA **578**, A14 (2015).
- [3] G. B. Field, Proceedings of the IRE **46**, 240 (1958).
- [4] A. L. Rudolph, Physics and Astronomy, Cal Poly Pomona .
- [5] F. Eisenhauer *et al.*, The Astrophysical Journal **597**, L121–L124 (2003).
- [6] E. H. D. Jr., **8121**, 812106 (2011).
- [7] P. Schneider, *Introduction and overview* (Springer Berlin Heidelberg, Berlin, Heidelberg, 2015), p. 6.
- [8] Golubov, O. *et al.*, AA **557**, A92 (2013).
- [9] H. Nyquist, Transactions of the American Institute of Electrical Engineers **47**, 617 (1928).
- [10] M. Honma and Y. Sofue, Publications of the Astronomical Society of Japan **49**, 453 (1997).
- [11] Y. Sofue, Galaxies **8** (2020).
- [12] V. Trimble, Contemporary Physics **29**, 373 (1988).
- [13] M. Milgrom, **270**, 365 (1983).
- [14] M. Milgrom, Canadian Journal of Physics **93**, 107–118 (2015).
- [15] Y. Huang *et al.*, Monthly Notices of the Royal Astronomical Society **463**, 2623 (2016).
- [16] K.-H. Chae, The Astrophysical Journal **941**, 55 (2022).
- [17] S. S. McGaugh, The Astrophysical Journal **609**, 652–666 (2004).
- [18] R. H. Sanders, **2**, 1 (1990).
- [19] A. Slosar, A. Melchiorri, and J. I. Silk, Physical Review D **72** (2005).
- [20] D. Clowe *et al.*, The Astrophysical Journal **648**, L109–L113 (2006).